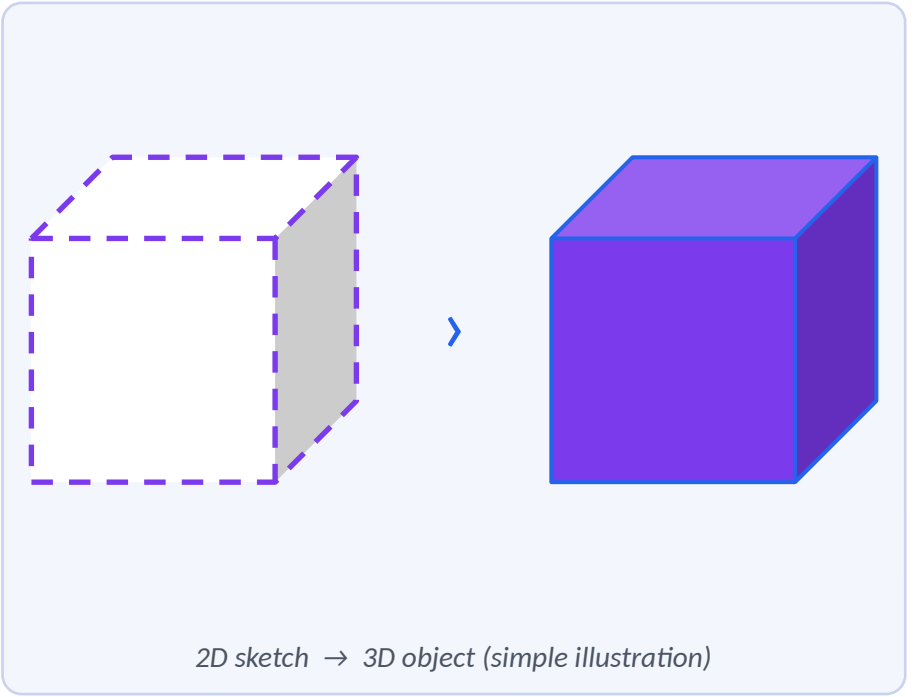
TechZephyr 4.0 | IIT Bhubaneswar

SKETCH2SHAPE AI

From 2D Sketches to 3D Models

A simple AI-based application that aims to convert sketches, uploaded images and text descriptions into 3D models.



2D sketch → 3D object (simple illustration)

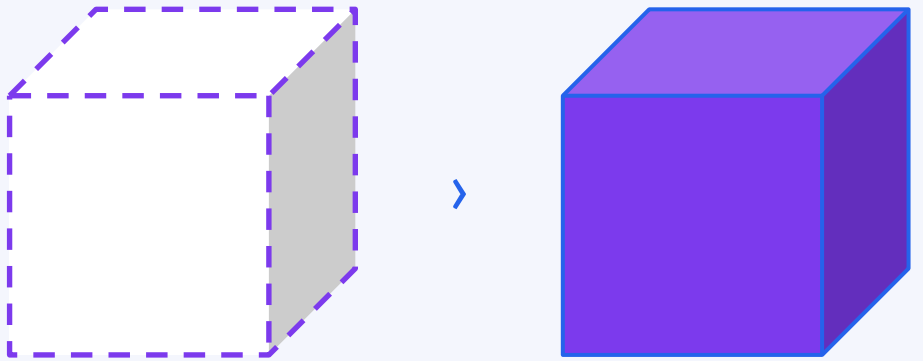
Team name: [ADD TEAM NAME]

Team members: [ADD NAMES]

Problem Statement

Making a 3D model from a simple drawing is not easy

- Creating 3D models manually can take a lot of time.
- Beginners may not know how to use professional 3D modelling software.
- Converting a simple drawing into a 3D object is challenging.
- Our project explores an easier way to generate basic 3D models from simple inputs.



2D sketch → 3D object (simple illustration)

Our Idea and Solution

The user picks one of three input methods

1. Sketch to 3D

Draw an object on a canvas and generate a 3D model.

[CONFIRM STATUS]

2. Upload Image to 3D

Upload an existing image and convert it into a 3D model.

[CONFIRM STATUS]

3. Text / Prompt to 3D

Describe an object in text. Only possible if a suitable text-to-3D model is added.

Planned

Note: TripoSR is an image-to-3D model, so sketch and image inputs use it. It cannot make 3D models from text by itself.

How Our Application Works

Website flow from home page to results

1. Home Page

- Project title
- Short description
- Explore and How It Works

2. Explore Page

- Sketch to 3D
- Upload Image to 3D
- Text / Prompt to 3D

3. Input

- User draws, uploads or types a prompt

4. 3D Result Page

- Shows the 3D model
- Rotate and zoom
- Custom height, width, colour + Run button
- GLB download

5. History + About Us

- Previously generated models
- Team, technologies, datasets, AI tools

Technology and Implementation

Only the tools we actually use

Python

Backend development

TripoSR

Image-to-3D reconstruction

PyTorch

Runs the AI model

Trimesh

Works with and checks 3D meshes

Gradio / Streamlit

App interface [CONFIRM ONE]

Basic process

Input Image / Sketch › **Image Processing** › **TripoSR** › **3D Mesh** › **GLB Export**

- TripoSR is an image-to-3D model, not a native text-to-3D model.
- Gemini was only an AI-assisted development tool that helped us plan and build. It is not the 3D generation model.
- Datasets: none directly used by our team [CONFIRM]. TripoSR is a pretrained model by Stability AI and Tripo AI.

Current Progress and Challenges

A student hackathon prototype; we show where we are honestly

Implemented

Implemented

- Basic website structure and input workflow
- Image-to-3D with TripoSR

In Progress

In Progress

- 3D preview (rotate and zoom)
- GLB export and file validation
- Better 3D quality and sketch match

Planned

Planned

- Text-to-3D model integration
- Better height, width and colour settings

Challenges

- Current 3D output quality needs improvement
- Models may not match the sketch
- GLB download needs testing and fixing

[INSERT APPLICATION SCREENSHOT]

[INSERT ACTUAL 3D OUTPUT]

Conclusion and Future Scope

Conclusion

- SKETCH2SHAPE AI aims to make basic 3D model generation easier for beginners.
- Users can give a sketch, image or text description through a simple interface.
- It is an attempt to connect AI-based reconstruction with a user-friendly application.

Future Improvements

- Improve 3D model quality
- Improve similarity between sketch and model
- Make GLB download reliable
- Improve custom height, width and colour settings
- Integrate a suitable text-to-3D model

Thank You